

Supplementary Information

Testing a proposed transcontinental great circle alignment: Land-constrained Monte Carlo baselines, geographic explanations, and localized monument divergence at the Memphis necropolis

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S1. Deep-time enrichment: early Holocene analysis

In the early Holocene (10,500–8,500 BP), mean enrichment within the 200 km corridor is 4.52× above background on the global p3k14c database, and this enrichment places the Alison circle above all nine tested Fertile Crescent-passing circles (range 0.88–2.71×), corresponding to $Z = 5.26$ above the mean of those nine circles. Seventeen of 19 near-circle sites in this epoch post-date the 10,200 BP PPNB break, consistent with the general pattern of Fertile Crescent agricultural expansion documented by Borrell et al. (2015).

However, the regional interpretation requires care. On the Near Eastern Radiocarbon Database (NERD, 11,072 dates), enrichment at this period is 1.10× — indistinguishable from the regional background. The 4.52× enrichment in p3k14c reflects the circle's multi-regional geometry connecting the Fertile Crescent, South America, and Easter Island on a single path relative to a global baseline, not evidence of a specific prehistoric corridor within the Near East. The circle's bearing at the center of the early Holocene Near Eastern site distribution is 88.1° (essentially east-west), while the natural settlement axis for the PPNB expansion is 14.3° (north-northeast to south-southwest) — nearly perpendicular to the circle. The early Holocene enrichment is a geometric coincidence, not a corridor signal.

S2. Circle family and geometric uniqueness

A search of 100,000 random great circles identified approximately 66 that pass within 200 km of all three anchor sites (Giza, Nazca, Easter Island) — roughly 1 in 1,515 random circles. On a 1° grid scan of 60 qualifying circles, the Alison circle ranks 5th by monument-settlement divergence D , at $D = 8.64$ (93.2nd percentile), when evaluated by triplet comparison. The top 4 poles are all within approximately 2° of the Alison pole. The best alternative anchor triplet (Athens + Persepolis + Varanasi) produces $D = 6.08$. The Alison circle is not the only geometric solution to connecting major archaeological sites along a single geodesic, but it ranks in the top 7% by archaeological signal strength.

Table S2-1. Circle family summary statistics: 60 qualifying great circles, ranked by monument-settlement divergence D. The Alison circle (pole: 59.682122°N, 138.646087°W) ranks 5th.

S3. Paleoclimate correlation details

Soreq Cave $\delta^{18}\text{O}$ speleothem records (Bar-Matthews et al. 2003), at a 500-year lag, correlate with the temporal divergence profile from p3k14c at $r = -0.77$, $p = 0.025$ (8 temporal bins). More negative $\delta^{18}\text{O}$ values — indicating wetter Eastern Mediterranean conditions — correlate with higher divergence at a 500-year lag. This is broadly consistent with the 3000–2500 BCE divergence peak coinciding with wetter-than-average conditions in the Levant before the 4.2-kiloyear aridification event.

After Benjamini-Hochberg correction across three lag values tested (0-year lag, 250-year lag, 500-year lag), the adjusted p value at the 500-year lag is approximately 0.075, which does not survive correction at $q = 0.05$. The 0-year and 250-year lags show weaker correlations ($r = -0.42$ and $r = -0.61$, respectively). The correlation is suggestive but not significant after multiple testing correction.

If real, this correlation would imply that climatic conditions affecting the Eastern Mediterranean at 500-year lead time influenced the pattern of monument versus settlement construction near the corridor. Climate may partly explain timing; it does not explain geometry. The small number of temporal bins ($n = 8$) makes this result fragile; a different binning scheme could alter it.

S4. Preservation test: extended trial details

The preservation test (main text §4.5) evaluated three conditions in detail.

Condition 1 — Baseline Pleiades Egypt data. 1,000 Monte Carlo trials using the Pleiades pre-2000 BCE Egypt subset as-is. $D = 12.83$. Monument $Z = 6.89$, settlement $Z = -5.94$.

Condition 2 — Adding 58 documented buried settlements. The 58 sites were drawn from the EES Delta Survey (Spencer 2024), Moeller (2016), and Kemp (2018), representing locations of known but archaeologically inaccessible buried settlements in the Nile Delta and floodplain. These sites were added to the Pleiades settlement class and the full Monte Carlo analysis was repeated. Short run (200 trials): $D = 13.49$. Long run (1,000 trials): $D = 13.10$. The increase in D under Condition 2 relative to Condition 1 occurs because the 58 documented buried sites cluster in the Delta (mean distance from the Alison great circle: 243 km, compared to 34 km for the pyramid fields). Adding these sites increases the settlement-depletion Z -score rather than offsetting it.

Condition 3 — Adding 350 Monte Carlo simulated settlements. 350 settlement sites were placed randomly within the Nile floodplain (25–31°N, 29–33°E) in each of 1,000 Monte Carlo outer-loop iterations, representing a conservative upper bound on the number of undocumented settlements that might hypothetically exist. For each outer-loop iteration, 1,000 inner-loop Monte Carlo trials were run to compute D. Results: mean D = 10.21, standard deviation = 0.72, minimum D = 8.03, maximum D = 12.18, fraction with D > 2: 100.0%. Under the most conservative possible assumption about missing settlement data, the divergence survives in every simulation.

The convergence of three conditions — each representing increasingly liberal assumptions about missing settlement data — supports the conclusion that differential preservation does not explain the monument-settlement divergence. The spatial logic is straightforward: the desert plateau location of the Memphis pyramid fields places them at a systematic geographic distance from the alluvial floodplain zones where settlement sites are both most numerous and most likely to be buried and underrepresented.